CRC Lab — Calculation Report

Generated by CRC Lab

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1 Inputs

Message	`hehe`
Data bits	`01101000011001010110100001100101` (32 bits)
Generator	`1011` (4 bits)
Appended zeros	3

2 Sender: polynomial long division

Augmented dividend (data + 3 zeros):

`01101000011001010110100001100101000`

Generator: `1011`. At each step a leading 1 is cancelled by XOR with the generator (⊕ marks such a step):

step	xor	register after step
0	—	01101000011001010110100001100101000
1	⊕	00110000011001010110100001100101000
2	⊕	00011100011001010110100001100101000
3	⊕	00001010011001010110100001100101000
4	⊕	00000001011001010110100001100101000
5	—	00000001011001010110100001100101000
6	—	00000001011001010110100001100101000
7	⊕	00000000000001010110100001100101000
8	—	00000000000001010110100001100101000
9	—	00000000000001010110100001100101000
10	—	00000000000001010110100001100101000
11	—	00000000000001010110100001100101000
12	—	00000000000001010110100001100101000
13	⊕	00000000000000011110100001100101000
14	—	00000000000000011110100001100101000
15	—	00000000000000011110100001100101000
16	⊕	00000000000000001011000011001010000
17	⊕	00000000000000000000000001100101000
18	—	00000000000000000000000001100101000
19	—	00000000000000000000000001100101000
20	—	00000000000000000000000001100101000
21	—	00000000000000000000000001100101000
22	—	00000000000000000000000001100101000
23	—	00000000000000000000000001100101000
24	—	00000000000000000000000001100101000
25	⊕	00000000000000000000000000111101000
26	⊕	00000000000000000000000000010001000
27	⊕	00000000000000000000000000000111000
28	—	00000000000000000000000000000111000
29	⊕	00000000000000000000000000000010100
30	⊕	00000000000000000000000000000000010
31	—	00000000000000000000000000000000010

3 Transmitted frame

CRC remainder: `010`. Frame = data + CRC:

`01101000011001010110100001100101010` (35 bits)

4 Receiver verification

Received frame: `01101000011001010110100001100101010`. Flipped bit positions: none. Dividing by `1011` gives remainder `000`.

VALID — no error

Remainder is all zero, so the frame arrived intact.